

Haseeb Asif

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Summary

Research oriented Computer Science undergraduate with experience in deep learning, generative AI, large language models, and agentic systems. Skilled in implementing and reproducing machine learning methods, conducting experimental evaluation, and developing AI applications. Currently exploring prompt injection resilience and AI safety for large language models, with research interests in post training, reinforcement learning, alignment, and trustworthy AI systems.

Education

FAST National University, B.S in Computer Science | Karachi, Pakistan | CGPA: 3.51/4 Aug 2023–Present
Relevant Coursework: Linear Algebra, Calculus, Multivariable Calculus, Probability & Statistics, Discrete Structures, Data Structures, Recommender Systems, Information Retrieval, Artificial Intelligence

Experience

Alexin AI, AI Research & Development | Remote (Bay Area, California) Aug 2026–Current

- Contributing to the memory layer design for an AI platform building persistent employee digital twins, grounded in a live knowledge graph with a four-tier, evaluation-gated memory hierarchy.
- Conducted a systematic review of 30+ papers on LLM agent memory (consolidation, forgetting, retrieval, multi-agent memory), screening for peer-review credibility and production relevance, and built a structured knowledge base organized around the system's open design questions.
- Present comparative technical findings at cross-team research reviews (e.g. prediction-error-based memory gating vs. rule-based scoring) to inform the memory layer's direction.

Algoma University, Mitacs Globalink Research Intern | Sault Ste. Marie, Canada July 2026–Current

- Conducting research on visual similarity based product recommendation systems, benchmarking embedding based architectures across CNN feature extractors (ResNet, VBPR) and vision language models (CLIP, FashionCLIP) for cross modal retrieval.
- Architecting a scalable experimentation pipeline on the Amazon Review Data 2018 (Clothing, Shoes & Jewelry) corpus to evaluate visual embedding techniques against retrieval accuracy and recommendation relevance metrics.
- Synthesizing a comparative literature review spanning CNN and VLM based recommendation architectures to surface state of the art techniques and identify gaps driving the next phase of experimentation.

Stevens Institute of Technology, Research Assistant | Remote (United States) Apr 2025–May 2025

- Developed a custom convolutional neural network in PyTorch with residual connections and dropout, achieving 95.2% classification accuracy on 5,000+ anonymized brain MRI scans for neurological disorder detection.
- Built a reproducible preprocessing pipeline in Python (NumPy, SciPy, Pillow) with intensity normalization, skull stripping, and affine transformations to standardize heterogeneous MRI datasets.
- Ran large-scale hyperparameter optimization with Ray Tune (distributed grid, random, and Bayesian search) and early stopping, reducing training time by 30%.
- Applied SHAP and LIME to produce voxel-level saliency maps for clinical validation and peer-reviewed research.

Research Projects

Qalb Post-Train — Urdu Math Reasoning via Verifiable-Reward RL 2026

- Extended Qalb-1.0-8B-Instruct (Llama-3.1-8B continued-pretrained on Urdu/English) past instruction-tuning with a verifiable-reward pipeline: rejection-sampling cold-start → SFT format primer → GRPO, teaching Urdu chain-of-thought reasoning with a machine-checkable numeric answer.
- Designed a rule-based reward (digit-normalized answer matching against GSM8K-Urdu) with no judge model, running the full pipeline under a limited compute budget on serverless GPUs.
- Engineered a resumable multi-stage training system across three isolated container images to resolve conflicting dependency requirements between the training, SFT, and GRPO stacks.

Shellsim — GRPO Terminal Agent Trained Against a Simulated Shell 2026

- Built a full RL training harness that replaces the real shell/Docker environment in terminal-agent RL with a large language model (Qwen-AgentWorld-35B-A3B) that simulates one, so a policy (Qwen2.5-3B-Instruct) can be trained via GRPO without ever touching a live container.
- Designed a controllable perturbation-based task-variant generator paired with declarative, machine-checkable verifiers, so reward stays fully rule-based and never depends on world-model or judge output.
- Engineered the end-to-end infrastructure (verl-based GRPO training, vLLM serving, Modal deployment) connecting the policy, the simulated environment, and reward computation into a single reproducible pipeline.

Honors & Awards _____

- **1st Place Overall & Best Social/Environmental Contribution** — EarthLink AI Feb 2026
- Selected for a highly competitive, fully funded **Mitacs Globalink Research Internship** Sep 2026
- **Qimam Fellowship** — Interview stage, 1 of 125 from 13,000+ applicants (~1%) July 2026
- **Dean’s List**, 4x 2023–2026
- **Best Team Award**, ACM Automation

Projects _____

- EarthLink AI** — *EnviroCast Global Engineering Outlook Hackathon* | **1st Place Overall** Feb 2026
- Architected an AI powered geospatial intelligence platform transforming natural language queries into environmental insights using Sentinel-2 satellite imagery, agentic LLM workflows, and dynamic tool orchestration via FastAPI and Google Gemini.
 - Engineered a multi-step analysis pipeline computing NDVI, land surface temperature (LST), built up intensity (NDBI), elevation, and vegetation metrics from multispectral satellite data.
 - Developed an interactive full-stack visualization interface using Next.js, React, and Mapbox GL for real time rendering of environmental hotspots and regional comparisons.

- Image2Depth** July 2025
- Implemented a Pix2Pix based image-to-image translation system in PyTorch to estimate monocular depth maps from single-view dashboard RGB images, using a U-Net generator with skip connections and a PatchGAN discriminator.
 - Optimized training with a composite adversarial and L1 reconstruction loss, achieving 23.8% reduction in MAE compared to baseline CNN regressors.
 - Automated hyperparameter tuning with Ray Tune (ASHA scheduler) and tracked experiments with Weights & Biases and MLflow, achieving 31% faster convergence.

- DeepFaceEdit** June 2025 – July 2025
- Architected a desktop application for real time human face synthesis and latent space exploration, leveraging StyleGAN2-ADA with a PyTorch backend.
 - Implemented a latent space manipulation engine enabling fine grained semantic editing of facial attributes through learned direction vectors in the W^+ latent space.
 - Engineered an image projection module to map real photographs into StyleGAN’s latent space, achieving 18% lower L2 reconstruction error compared to baseline pipelines.

- Advance Lane Detection (Classical CV)** Apr 2025 – May 2025
- Developed a complete lane detection pipeline from scratch using OpenCV and NumPy, tailored for ADAS style real time video feeds and static road imagery.
 - Implemented camera calibration, distortion correction, and perspective transform, with binary thresholding and sliding window search to fit lane polynomials.
 - Calculated lane curvature and vehicle offset in real world units, validating the system with 94.2% pixel-level accuracy under challenging driving conditions.

Skills _____

Programming Languages: Python, C, C++, JavaScript | ML Frameworks: Scikit-learn, TensorFlow, PyTorch, PyTorch Lightning, JAX, ONNX, MLflow, Ray, Apache Airflow, Weights & Biases, TorchServe | Computer Vision: OpenCV, CARLA, Matplotlib, Plotly | Geospatial & Remote Sensing: Google Earth Engine, Sentinel-2 MSI, NDVI/NDBI/LST Analysis, Mapbox GL, GeoJSON | Data Processing: Pandas, NumPy | LLM & RAG: LangChain, Pinecone, FAISS, ChromaDB, CrewAI, Model Context Protocol (MCP) | Web & Automation: FastAPI, Streamlit, Selenium, Postman, Discord.py | Frontend Development: Next.js, HTML, CSS | DevOps: Git, GitHub Actions, Docker | Tools: VS Code, CursorAI, Google Colab, Hugging Face Spaces | Databases: MongoDB | Soft Skills: Research-Oriented Thinking, Problem Solving, Scientific Communication, Team Collaboration, Self Directed Learning

Certificates _____

- ML Summer School 2025** — Cohere 2025
- Participated in a curated program focused on foundational and advanced machine learning concepts.
- Generative Adversarial Networks (GANs) Specialization** — DeepLearning.AI 2025
- Gained hands-on experience building and training GAN models for image generation tasks.
- Ray Fundamentals Accreditation** — AnyScale 2025
- Learned fundamentals of distributed computing and ML training using Ray.
- Deep Learning with PyTorch: Siamese Network** — Coursera 2025
- Completed a course on Siamese Networks using PyTorch for similarity learning tasks.